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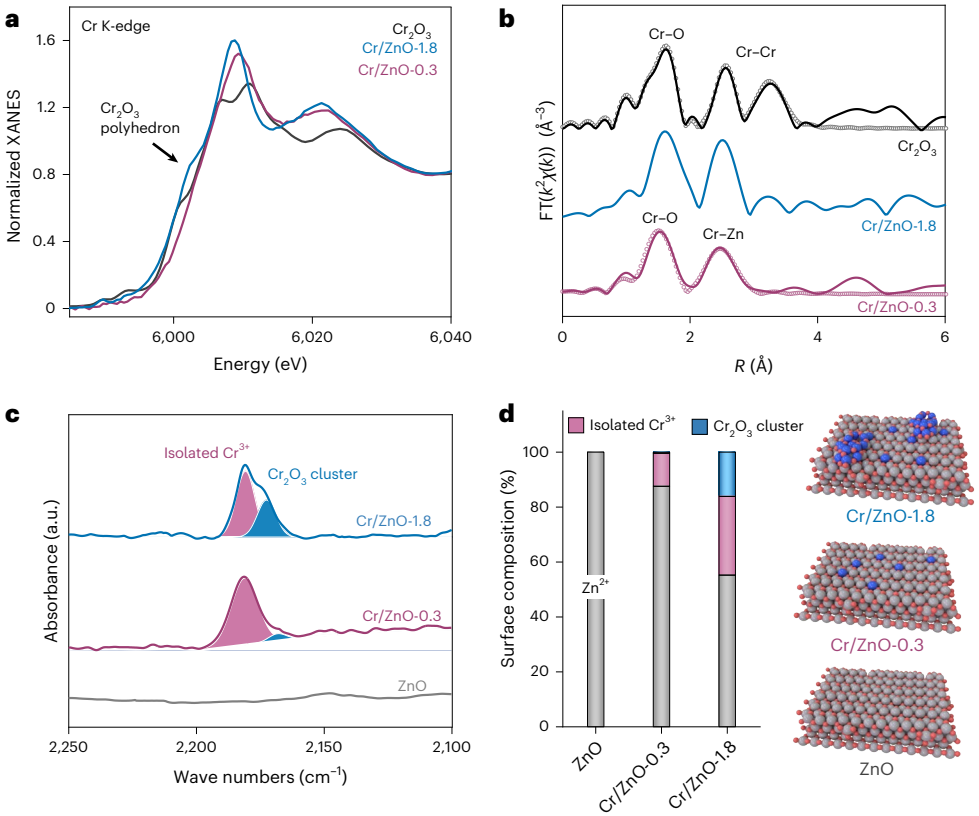


Fig. 1 | The structure of Cr/ZnO. a, XANES of used Cr/ZnO-*x* catalysts and Cr_2O_3 after syngas conversion reaction. **b**, Fourier transforms and fitting of the k^2-weighted Cr K-edge EXAFS spectra of Cr/ZnO-*x* catalysts and Cr_2O_3. The vertical axis, FT($k^2\chi(k)$), represents the magnitude of the Fourier transform of the k^2-weighted EXAFS function $\chi(k)$, and *R* represents the radial distance from the absorbing Cr atom. **c**, In situ CO-IR at −100 °C for Cr/ZnO-*x* and ZnO catalysts. **d**, Surface composition and schematic diagram of Cr/ZnO-*x* catalysts.

oxygen sites: via removal of a surface oxygen on ZnO, an interfacial oxygen atom at Cr_{iso}–O–Zn or an interfacial oxygen at Cr_2O_3 cluster–O–Zn boundary (Fig. 2a).

Preferential locations of O_v at Cr/ZnO surface

O_v typically forms under reducing conditions by the removal of surface lattice oxygen, a process accompanied by reduction of neighbouring cations and redistribution of electrons. In our in situ XPS experiments under reductive atmosphere, the maximum shifts of Cr 2*p* and Zn 2*p* peaks were ~0.8 eV and ~0.2 eV, respectively, and the O 1*s* intensity at ~530 eV decreased slightly, consistent with electron enrichment of Cr and Zn and some loss of lattice oxygen[33] (Supplementary Figs. 16–18). While these XPS signatures indicate O_v generation, they provide only indirect evidence and cannot identify where on the surface the vacancies form or their quantity. We therefore used a suite of complementary techniques, integrating experimental and computational approaches, to localize and quantify O_v formation at the various Cr/ZnO surface motifs. In particular, we sought multiple lines of evidence for O_v formation: a decrease in surface metal–oxygen coordination number, a measurable loss of surface oxygen content and detection of oxygen being removed as H_2O/CO_2 upon reduction.

H_2 temperature-programmed reduction (TPR) provided an initial indication of O_v formation. Cr/ZnO-0.3 with exclusively isolated Cr^{3+} showed a distinct reduction peak around 350 °C (Supplementary Fig. 19). Cr/ZnO-1.8 displayed a sharp peak at ~350 °C and a broad one above 400 °C, suggesting the presence of both easily reducible oxygen (probably near isolated Cr^{3+}) and more resistant oxygen (possibly at Cr_2O_3 cluster–O–Zn interface). By contrast, pure ZnO required temperatures above 590 °C for any noticeable reduction. Importantly, cycling the samples between reduction (TPR) and re-oxidation (temperature-programmed oxidation) demonstrated that oxygen removed during H_2 exposure could be replenished upon re-oxidation, with minimal loss of capacity over multiple cycles (Supplementary Fig. 20). This suggests that O_v formation on Cr/ZnO-*x* is reversible and highlights the promotional effect of Cr^{3+} and the dependence of O_v formation on Cr^{3+} speciation.

While conventional XAS averages metal–O coordination over bulk and surface, our aim was to specifically probe the Zn–O coordination and the relation between O_v and Zn atoms at the surface under redox cycles. We developed a TKA coupled with EDXAS and mass spectrometry (TKA–EDXAS–MS) to monitor, in real time, the changes in Zn K-edge EDXAS during alternating 20% H_2/Ar and air flows every few minutes at 400 °C. EDXAS was collected at 1 Hz enabling a high time resolution for dynamic studies (Supplementary Table 7). The rapid H_2/air redox perturbations scarcely affect lattice oxygen in bulk, thereby enhancing surface sensitivity of EDXAS (Supplementary Fig. 21). We therefore prioritized operando time-resolved measurements at the Zn K-edge because the primary structural process under investigation is the evolution of O_v within the ZnO lattice[34]. Because Cr is highly dispersed on ZnO, directly resolving O_v formation at the Cr–O_v–Zn interface is intrinsically challenging owing to the overwhelming contribution from the bulk ZnO lattice signal.

Upon H_2 exposure at 400 °C, we observed a clear decrease in the Zn K-edge white-line intensity (9,660–9,675 eV) for Cr/ZnO-1.8 within 350 s (Fig. 2b and Supplementary Fig. 22). Concurrently, the amplitude of the first-shell Zn–O peak in the Fourier-transformed EXAFS declined slightly (Fig. 2c), directly indicating a loss of oxygen neighbours around Zn atoms at the surface. This matches density functional theory (DFT) simulations of oxygen removal around Cr_1O_1/ZnO (Supplementary Fig. 23). Simultaneous mass spectroscopic analysis